

Cybersecurity Glossary

A

Advanced persistent threat (APT): An instance when a threat actor maintains unauthorized access to a system for an extended period of time

AI bias: Risks involved when bias is present in artificial intelligence (AI) data, potentially leading to inaccurate, unfair, or unreliable outputs

Analysis: The investigation and validation of alerts

Anomaly-based analysis: A detection method that identifies abnormal behavior

Array: A data type that stores data in a comma-separated ordered list

Artificial intelligence (AI): Computer programs designed to perform cognitive tasks typically associated with human intelligence, such as learning, problem-solving, and understanding language

B

Broken chain of custody: Inconsistencies in the collection and logging of evidence in the chain of custody

Business continuity plan (BCP): A document that outlines the procedures to sustain business operations during and after a significant disruption

C

Chain of custody: The process of documenting evidence possession and control during an incident lifecycle

Command and control (C2): The techniques used by malicious actors to maintain communications with compromised systems

Command-line interface (CLI): A text-based user interface that uses commands to interact with the computer

Common Event Format (CEF): A log format that uses key-value pairs to structure data and identify fields and their corresponding values

Computer security incident response teams (CSIRT) : A specialized group of security professionals that are trained in incident management and response

Configuration file: A file used to configure the settings of an application

Containment: The act of limiting and preventing additional damage caused by an Incident

Crowdsourcing: The practice of gathering information using public collaboration

Confidential data: Data that often has limits on the number of people who have access to it

Context (in prompting): The necessary details provided to an artificial intelligence (AI) tool to help it understand what you need from it.

D

Data controller: A person that determines the procedure and purpose for processing data

Data exfiltration: Unauthorized transmission of data from a system

Data packet: A basic unit of information that travels from one device to another within a network

Data processor: A person that is responsible for processing data on behalf of the data controller

Data protection officer (DPO): An individual that is responsible for monitoring the compliance of an organization's data protection procedures

Detection: The prompt discovery of security events

Documentation: Any form of recorded content that is used for a specific purpose

Disaster recovery plan: A plan that allows an organization's security team to outline the steps needed to minimize the impact of a security incident

E

Endpoint: Any device connected on a network

Endpoint detection and response (EDR): An application that monitors an endpoint for malicious activity

Eradication: The complete removal of the incident elements from all affected systems

Escalation policy: A set of actions that outlines who should be notified when an incident alert occurs and how that incident should be handled

Evaluate (in prompting): The step in the TCREI framework where you review the output from an artificial intelligence (AI) tool to determine if it meets your needs and Expectations

Event: An observable occurrence on a network, system, or device

F

False negative: A state where the presence of a threat is not detected

False positive: An alert that incorrectly detects the presence of a threat

Final report: Documentation that provides a comprehensive review of an incident

G

Generative AI (Gen AI): A type of artificial intelligence (AI) that is capable of creating new content, such as text, images, code, or other media, in response to prompts

H

Honeypot: A system or resource created as a decoy vulnerable to attacks with the purpose of attracting potential intruders

Host-based intrusion detection system (HIDS): An application that monitors the activity of the host on which it's installed

Human-in-the-loop approach: A method of using artificial intelligence (AI) responsibly that combines the strengths of machine intelligence with human oversight, involvement, and validation to train, use, verify, and refine AI results

I

Incident : An occurrence that actually or imminently jeopardizes , without lawful authority, the confidentiality , integrity or availability of information or an information system ; or constitutes a violation or imminent threat of violation of law , security policies , security procedures or acceptable use policies

Incident handler's journal : A form of documentation used in incident response

Incident response plan: A document that outlines the procedures to take in each step of incident response

Indicators of attack (IoA): The series of observed events that indicate a real-time incident

Indicators of compromise (IoC): Observable evidence that suggests signs of a potential security incident

Internet Protocol (IP): A set of standards used for routing and addressing data packets as they travel between devices on a network

Intrusion detection system (IDS): An application that monitors system activity and alerts on possible intrusions

Intrusion prevention system (IPS): An application that monitors system activity for intrusive activity and takes action to stop the activity

Improper usage: An incident type that occurs when an employee of an organization violates the organization's acceptable use policies

Improper escalation : The process of identifying a potential security incident , triaging it, and handling it off to be a more experienced team member

Iterate (in prompting): The process in the TCREI framework of refining and improving artificial intelligence (AI) prompts and outputs through repeated cycles of testing and

adjustments to achieve desired results

K

Key-value pair: A set of data that represents two linked items: a key, and its corresponding value

L

Lessons learned meeting: A meeting that includes all involved parties after a major Incident

Log analysis: The process of examining logs to identify events of interest

Log management: The process of collecting, storing, analyzing, and disposing of log Data

Logging: The recording of events occurring on computer systems and networks

M

Malware infection : An incident type that occurs when malicious software designed to disrupt a system infiltrates an organization's computers or network

Media Access Control (MAC) Address: A unique alphanumeric identifier that is assigned to each physical device on a network

N

National Institute of Standards and Technology (NIST) Incident Response

Lifecycle: A framework for incident response consisting of four phases: Preparation; Detection and Analysis; Containment, Eradication, and Recovery; and Post-incident activity

Network-based intrusion detection system (NIDS): An application that collects and

monitors network traffic and network data

Network data: The data that's transmitted between devices on a network

Network protocol analyzer (packet sniffer): A tool designed to capture and analyze data traffic within a network

Network traffic: The amount of data that moves across a network

Network Interface Card (NIC): Hardware that connects computers to a network

O

Object: A data type that stores data in a comma-separated list of key-value pairs

Open-source intelligence (OSINT): The collection and analysis of information from publicly available sources to generate usable intelligence

OWASP Top 10: A globally recognized standard awareness document that lists the top 10 most critical security risks to web applications

P

Packet capture (p-cap): A file containing data packets intercepted from an interface or network

Packet sniffing: The practice of capturing and inspecting data packets across a network

Playbook: A manual that provides details about any operational action

Post-incident activity: The process of reviewing an incident to identify areas for improvement during incident handling

Private data: Information that should be kept from the public

Prompt: The input (which can be text, images, or other data) you provide to an artificial intelligence (AI) model to elicit a specific response or generate new content

Public data: Data that is already accessible to the public and poses a minimal risk to

the organization if viewed or shared by others

Q

R

Recovery: The process of returning affected systems back to normal operations

Resilience: The ability to prepare for, respond to, and recover from disruptions

Root user (or superuser): A user with elevated privileges to modify the system

S

Search Processing Language (SPL): Splunk's query language

Security information and event management (SIEM): An application that collects and analyzes log data to monitor critical activities in an organization

Security operations center (SOC): An organizational unit dedicated to monitoring networks, systems, and devices for security threats or attacks

Security orchestration, automation, and response (SOAR): A collection of applications, tools, and workflows that uses automation to respond to security events

Signature: A pattern that is associated with malicious activity

Signature analysis: A detection method used to find events interest

Stakeholder: An individual or a group that has an interest in any decision or activity of an organization

Standards: References that inform how to set policies

STAR method: A storytelling framework that helps you communicate your experiences and skills in a clear, concise, and compelling way to help an interviewer understand exactly how you successfully managed certain situations; stands for Situation, Task, Action, and Result; you'll learn more about this method in the last course of the

Certificate

Sudo: A command that temporarily grants elevated permissions to specific users

Suricata: An open-source intrusion detection system and intrusion prevention system

T

Task (in prompting): The specific instruction or goal you give to an artificial intelligence (AI) model, often including details about the desired persona and format for the output

tcpdump: A command-line network protocol analyzer

TCREI framework: A practical framework for writing effective prompts for generative artificial intelligence (AI) tools, which stands for Task, Context, References, Evaluate, and Iterate

Telemetry: The collection and transmission of data for analysis

Threat hunting: The proactive search for threats on a network

Threat intelligence: Evidence-based threat information that provides context about existing or emerging threats

Triage: The prioritizing of incidents according to their level of importance or urgency

True negative: A state where there is no detection of malicious activity

True positive An alert that correctly detects the presence of an attack

U

Unauthorized access: An incident type that occurs when an individual gains digital or physical access to a system or an application without permission

V

Visual dashboard: A way of displaying various types of data quickly in one place

W

Wildcard: A special character that can be substituted with any other character

Wireshark: An open-source network protocol analyzer

Y

YARA-L: A computer language used to create rules for searching through ingested log data

Z

Zero-day: An exploit that was previously unknown